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Dynamic real-time optimization to mitigate critical state effects in expert-controlled SAG mills

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ABSTRACT

Semi-autogenous (SAG) mills are critical components in mineral processing, widely used for size reduction to liberate valuable minerals in ore. These mills account for a significant portion of energy consumption and operational costs in mineral concentrator plants, making their optimization a key factor for improving overall process efficiency. Traditional expert control systems used in SAG mill operations lack predictive capabilities, which can result in critical operational states requiring a halt in the process and significantly affecting productivity.

This work introduces a dynamic real-time optimization (D-RTO) strategy to mitigate the impact of critical states, such as overload or high pressure, in SAG mills controlled by expert systems. The proposed supervisory layer dynamically adjusts the limits of controlled variables based on predictive capabilities provided by a digital twin. Simulation results demonstrate significant performance improvements, including reductions in total tonnage loss of up to 57%, decreases in critical state duration by up to 33%, and increases in average throughput by up to 3.5%. These results underline the effectiveness of the D-RTO strategy, particularly under dynamic and variable conditions. Future research will expand the framework to include power consumption considerations for comprehensive economic optimization.

1. Introduction

Comminution in mineral processing plants involves a sequence of crushing and grinding operations to reduce the size of ore particles, enabling the liberation of valuable minerals. Grinding circuits play a pivotal role in mineral processing by reducing ore particle sizes to enhance downstream separation and mineral liberation processes (Bouajila, Bartolacci, Kock, Cayouette, & Coté, 2000; Salazar, Valdés-González, & Cubillos, 2010; Taylor, Skuse, Blackburn, & Greenwood, 2020). However, these circuits are recognized as some of the most energy-intensive operations in the mining industry, typically accounting for approximately one-third of a mine's total energy usage (Craig, 2012; Lestage, Pomerleau, & Hodouin, 2002; Wei & Craig, 2009). The increasing costs associated with energy and the demand for more sustainable operations have intensified the need for advanced control strategies that improve operational efficiency while maintaining product quality (Perez-Garcia, Bouchard, & Poulin, 2021).

Grinding typically occurs in tumbling or stirred mills, where ore is reduced through the combined mechanisms of impact, attrition, and abrasion caused by the motion of unconnected grinding media, such as steel rods, steel or ceramic balls, or coarse ore pebbles (Liao, Xu, Chen, Wang, & She, 2024). Autogenous Grinding (AG) mills use the ore itself as grinding media. However, due to the breakage characteristics of hard ore, AG milling is often unsuitable. Instead, Semi-Autogenous Grinding (SAG) mills are employed, where steel balls are added to assist with ore breakage. SAG mills represent some of the highest throughput grinding circuits in the mining industry, facilitating mineral liberation that is critical for downstream concentration and extraction processes. These mills offer several advantages, including simplified process flows and lower steel consumption compared to traditional grinding methods. As a result, SAG mills have become the preferred design choice for modern mineral processing plants, with new installations rarely considering alternative configurations (Craig, 2012; Powell et al., 2023; Salazar et al., 2010).

Over the years, significant efforts have been directed at improving the performance of SAG mills, focusing on maximizing throughput, enhancing energy efficiency, and optimizing the performance of the comminution process for a variety of ore types (Bouajila et al., 2000; Taylor et al., 2020). These advancements highlight the essential role

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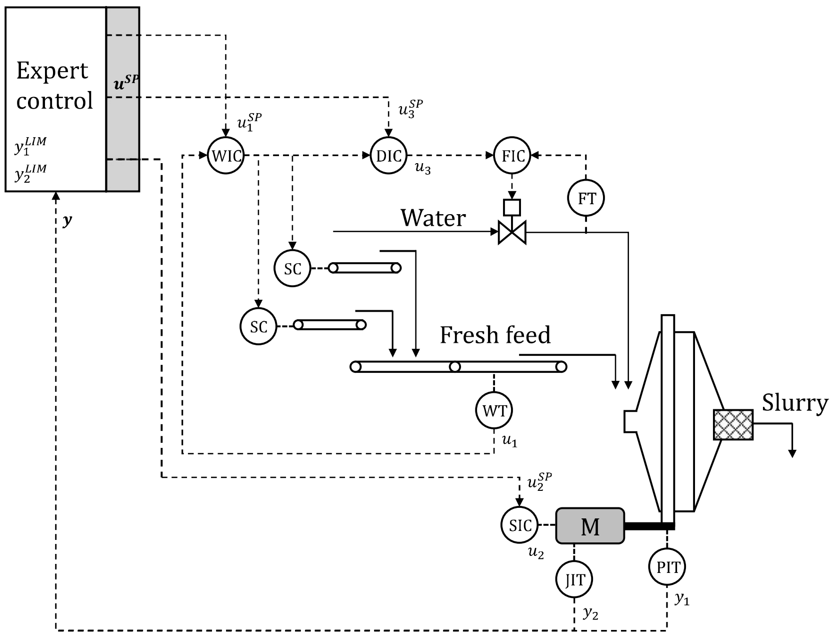

Fig. 1. Classic SAG mill process and control scheme. SAG mill is fed with water and fresh ore via dedicated controllers. PIT corresponds to the bearing pressure transmitter of the mill and JIT to the power transmitter of the motor. These values are given to the expert control with their respective desired limits and to return the setpoints $\boldsymbol{u}^{SP}$ of the MVs. The mass flow of fresh ore feed setpoint u_1^{SP} is given to the mass flow controller WIC, the setpoint of speed of rotation of the mill u_2^{SP} to the speed controller of the motor SIC, and the density setpoint u_3^{SP}, calculated as solid percentage feed, is given to the density controller DIC. The density controller works in cascade with the water flow controller FIC and transmitter FT, and uses the WIC values for its calculation.

of SAG mills in achieving economically viable and sustainable mineral processing operations. However, SAG mill circuits are challenging to control due to significant external disturbances, inadequate process models, nonlinearities, and the lack of key process variable measurements (Craig, 2012; Matthews & Craig, 2013). Moreover, maintaining independent control over critical parameters such as product particle size, throughput, and product density is complicated by the inability to separately manage the amount, size, and hardness of the grinding medium within the mill. Variations in hardness and size create substantial disturbances, requiring feedback control to ensure efficient plant operation (Adel, Ulsoy, & Sastry, 1983).

Traditional control approaches, including Proportional-Integral-Derivative (PID) controllers, often fail to effectively manage the complex dynamics of these systems. These approaches struggle with external disturbances, such as variations in ore hardness and feed rate, and typically require substantial manual intervention from operators (Chen, Li, & Fei, 2008; Perez-Garcia et al., 2021). Expert control systems, introduced as a solution to overcome these challenges, leverage operator knowledge and predefined rules to dynamically adjust setpoints and manage the process more effectively. While expert systems have demonstrated improvements in stability and throughput, their performance is constrained by their reliance on static rules and heuristics. These systems lack the capability to adapt optimally to evolving operating conditions or to incorporate economic objectives into the control framework explicitly (Chen et al., 2008; Le Roux & Craig, 2019). For instance, while they can stabilize operations and reduce operator dependency, they do not account for trade-offs between energy efficiency and production throughput, limiting their potential for overall plant optimization (Le Roux & Craig, 2019).

Advanced model-based control strategies for both SAG (e.g., Garrido and Sbarbaro (2009), Salazar et al. (2010), Salazar, Valdés-González, Vyhmesiter, and Cubillos (2014), Yutronic and Toro (2010)), ball mills (e.g., Chen et al. (2008), Chen, Zhai, Li, and Li (2007), Ramasamy, Narayanan, and Rao (2005), Yang, Li, Chen, and Li (2010)), and integrated milling circuits (e.g., Botha, Le Roux, and Craig (2018), Coetzee, Craig, and Kerrigan (2010), Duarte, Castillo, Sepúlveda, Contreras, Giménez, and Castelli (2002), Karelovic, Razzetto, and Cipriano (2013), Nieto, Olivares, Gatica, Ramos, and Olmos (2009)), have been reported in the literature. Additionally, emerging methods, such as reinforcement learning, show strong potential in simulation environments (e.g., Guo, Wang, and Zhang (2019), Hallén, Åstrand, Sikström, and Servin (2019), Wei, Ren, Cheng, and Yan (2023)). Real-time optimization (RTO) strategies are reported much less frequently in the context of SAG mill control. RTO has demonstrated strong potential for optimizing complex, nonlinear processes under constraints (Darby, Nikolaou, Jones, & Nicholson, 2011). For instance, Lestage et al. (2002) implemented a linear programming-based RTO scheme for a simulated rod-mill ball-mill grinding circuit. For instance, Lestage et al. (2002) implemented a linear programming (LP)-based RTO scheme for a simulated rod-mill–ball-mill grinding circuit, aiming to maximize throughput while maintaining product fineness and respecting operating constraints. Their strategy leveraged a steady-state LP optimizer combined with bias correction for mild nonlinearities and demonstrated effective performance under realistic disturbances. However, transient constraint handling was limited and required additional control layers. The authors emphasized the importance of careful model identification and robust safety mechanisms to address abnormal conditions.

Dynamic RTO (D-RTO) strategies integrate dynamic process models with economic performance objectives to determine optimal setpoints in real time, continuously adapting to process variations and operational disturbances. D-RTO schemes surpass the limitations of steady-state RTO by capturing the transient dynamics of the system, enabling more accurate and proactive decision-making under non-stationary operating conditions. This makes it particularly valuable in processes where disturbances, such as unmeasured changes in ore hardness during mineral grinding, can significantly impact operational efficiency. Additionally, the rise of digital twins can greatly enhance D-RTO functionalities. Digital twins are high-fidelity, real-time virtual representations of physical systems that replicate both the current state and predicted future behavior of the process. They enable predictive analysis and scenario evaluation, facilitating the simulation and optimization of control actions prior to their implementation (Ghasemi et al., 2024; Melesse, Di Pasquale, & Riemma, 2020; Quintanilla, Fernández, Mancilla, Rojas, & Navia, 2025).

This work proposes a D-RTO strategy that leverages a digital twin from Quintanilla et al. (2025) to proactively adjust operational limits based on predicted behavior with minimal disruption to the existing control infrastructure. This strategy serves as an upper control layer to the well-established expert control system, which has been finely tuned over years of operational experience at the plant. The D-RTO layer can proactively modify the operational limits by using predictive capabilities to prevent the SAG mill from entering critical operational states. The proposed strategy is benchmarked against traditional expert control systems (i.e. without D-RTO) to highlight its superior performance. The D-RTO layer is not intended to replace the existing expert system, but rather to complement it by addressing its known limitations under overload conditions. In normal operation, the expert controller remains fully responsible for stabilization whenever possible. The D-RTO is only activated when a critical condition related to inventory build-up is detected, dynamically adjusting the controlled variable limits. This hierarchical design ensures that the benefits of the expert system are preserved for nominal conditions, while the D-RTO intervenes selectively in adverse scenarios. To the best of our knowledge, no previous studies have implemented a D-RTO layer on top of an existing expert control system in SAG milling operations.

This paper is organized as follows: Section 2 provides a detailed description of the SAG mill process, including key operating variables and their role in system performance. Section 3 introduces the proposed D-RTO system, describing its integration with the existing expert control layer. Section 4 presents the results and discussion, comparing the performance of the proposed D-RTO strategy against traditional expert systems under different operating scenarios. Finally, the conclusions summarize the key findings and contributions of this work in Section 5.

2. Process description and operating variables

The primary goal of the comminution process is to produce a product with consistent fineness at the highest possible throughput (Le Roux & Craig, 2019; Wills & Finch, 2016). Traditionally, the control objectives for a milling circuit include the improvement of the quality of the product by increasing the fineness of the grind and decreasing product size fluctuation, maximizing throughput, minimizing consumption of grinding media, and/or minimizing power consumption (Craig & MacLeod, 1995; Matthews & Craig, 2013).

The system under study (Fig. 1) consists of a SAG mill integrated with a sieve classifier for the outflow, controlled by an expert control and a regulatory control layer composed of PID controllers. The controlled variables (CVs) of the expert control are:

- Pressure of the bearings of the mill (y_1): Proportional to the inventory inside the mill.
- Power draw on the motor (y_2).

The manipulated variables (MVs) of the expert control are:

- The setpoint of ore feed tonnage to the mill (u_1^{SP}).
- The setpoint of the speed of rotation of the mill, expressed as a percentage of its critical value (u_2^{SP}).
- The setpoint of solid percentage, which calculates a water addition according to the actual setpoint of ore feed (u_3^{SP}).

The setpoints $\boldsymbol{u}^{SP} = [u_1^{SP}, u_2^{SP}, u_3^{SP}]$ are defined by the expert control and passed to the PID controllers in the regulatory control layer. These PID controllers aim to bring the respective process variables (PVs) $\boldsymbol{u} = [u_1, u_2, u_3]$ as close as possible to their setpoints $\boldsymbol{u}^{SP}$.

The expert control seeks to maximize throughput (ore feed setpoint) while maintaining stable operation and avoiding critical conditions such as high pressure or excessive power draw. Operators or metallurgical supervisors determine the maximum and minimum allowed setpoints for the MVs ($\boldsymbol{u}^{SP}|_{\max}^{\min}$), based on factors like economic objectives, equipment limits, and process stability. Additionally, high limits for the CVs ($\boldsymbol{y}^{LIM} = [y_1^{LIM}, y_2^{LIM}]$) are defined to ensure equipment protection and process stability. During operation, the expert control prioritizes the adjustment of solid percentage (u_3^{SP}) as the primary MV to influence $\boldsymbol{y}$. Adjustments to rotation speed (u_2^{SP}) are secondary, while changes to feed tonnage (u_1^{SP}) are a last resort, given their negative impact on economic objectives. Reducing ore feed tonnage quickly influences CVs but lowers overall plant throughput (Festa, Cornejo, Orrante, Alanis, & Gutierrez, 2009; Metzner et al., 2009; van Zyl, Paquot, Metzner, Fouche, & Gomez, 2013).

However, while the expert control system reacts to changes in pressure and power, it lacks predictive capabilities. For instance, under increasing ore hardness, reactive adjustments alone are insufficient to either prevent critical conditions from developing, or control the SAG mill during critical conditions if waiting times are active. Critical conditions refer to operational states where the accumulation of material within the mill escalates uncontrollably, leading to reduced milling efficiency. As ore hardness increases, it becomes progressively harder to break down the particles, causing material to accumulate. This phenomenon is exacerbated by the sieve's inability to discharge oversized material effectively. This leads to a cumulative effect, where the accumulation of material worsens over time. As a result, there is a significant reduction in milling capacity, increased energy consumption, and an overload of ore inside the mill. These limitations highlight the need for dynamic optimization to anticipate and address critical conditions proactively.

3. Methodology: Dynamic real-time optimization strategy (D-RTO)

In this section, we present the development of a D-RTO strategy that leverages a digital twin from Quintanilla et al. (2025) as a prediction model. This digital twin was developed with industrial data from a Chilean company and integrates the expert control (Section 3.1.1) and regulatory PID control (Section 3.1.2) layers with a predictive model based on RNNs (Section 3.1.3). A disturbance estimation module (Section 3.1.4) was also included to account for the plant-model mismatch between the prediction model estimation and the actual process measurements. An optimization module (Section 3.2) evaluates future scenarios based on predicted process dynamics and identifies the best adjustments to the high limits of controlled variables, such as mill pressure, to balance maximum tonnage throughput with stable operation. This predictive capability allows for dynamic and proactive adjustments, preventing the occurrence of critical conditions while maximizing operational efficiency.

The proposed D-RTO strategy ensures improved performance while reducing the occurrence and severity of critical states. The D-RTO predicts future states of the SAG mill, dynamically adjusting the CV limits ($\boldsymbol{y}^{LIM}$) to mitigate critical states. This proactive approach reduces variability in the controlled variables and minimizes economic losses caused by unnecessary reductions in tonnage throughput.

The anticipated reduction in critical conditions directly enhances overall performance in several ways. First, it helps prevent material accumulation within the mill, avoiding cascading effects that would otherwise degrade milling efficiency and reduce throughput. Second, by promoting a more stable process, the system ensures extended residence time, allowing for more effective grinding and improved particle classification at the sieve, ultimately leading to better product quality. Finally, the system reduces waiting times by limiting interruptions caused by manual adjustments or system resets, supporting more continuous and reliable operation.

3.1. Prediction model ($\mathcal{H}$)

The prediction model $\mathcal{H}$ consists of $\mathcal{H}_{\text{SAG}}$ and $\mathcal{H}_{\text{dist}}$, as shown in Fig. 2. The SAG mill process and control models, represented by $\mathcal{H}_{\text{SAG}}$, are composed of three models: (1) a model of the expert control system $\mathcal{H}_{\text{SAG}}^{\text{EXP}}$, (2) a model of the regulatory control layer via a PID control

Table 1
Operational states categorized as Critical or High Pressure. Based on specific rules and fuzzy logic, each state triggers a setpoint adjustment for one or more manipulated variables MVs, where increases and decreases cannot occur simultaneously. Note: The color highlights in this table correspond to the results shown in Fig. 5, as well as S7 and S9 in the supplementary material.

Category	Operational state name	Possible setpoint changes					
		u_3^{SP}		u_2^{SP}		u_1^{SP}	
		dec	*inc*	*dec*	*inc*	*dec*	*inc*
Critical	Overload	✓			✓	✓	
	Power Critical		✓	✓		✓	
	Pressure Critical	✓			✓	✓	
High Pressure	Pressure High and Increasing, Power High	✓				✓	
	Pressure High and Increasing, Power not High	✓		✓	✓	✓	
	Pressure High and not Increasing, Power High	✓	✓		✓	✓	
	Pressure High and not Increasing, Power not High	✓			✓	✓	✓

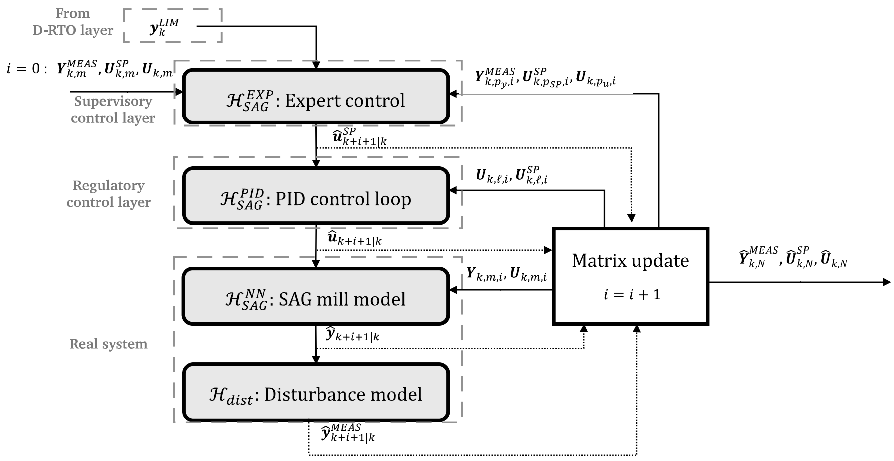

Fig. 2. Diagram of the proposed hierarchical control (Skogestad, 2004) with prediction model $\mathcal{H}$. Historical and/or predicted data is given to the expert control model with y_k^{LIM} to simulate the values $\hat{u}_{k+i+1}^{SP}$ for the next discrete time. The PID control model then uses all previous data to simulate the dynamics of each control loop and determine the predicted value of $\hat{u}_{k+i+1}$. Then, a model of the SAG mill process will use the data and predictions to determine the predicted values of $\hat{y}_{k+i+1}$. The disturbance model provides the prediction for the measured value $\hat{y}_{k+i+1}^{\mathrm{MEAS}}$. The above is done for the complete simulation prediction horizon from $i = 0, \ldots, (N-1)$. The blue boxes represent the digital twin developed in Quintanilla et al. (2025), while the orange box is the new addition of a disturbance model described in Section 3.1.4.

$\mathcal{H}_{\mathrm{SAG}}^{\mathrm{PID}}$, and (3) a model of the process for the SAG mill using RNNs $\mathcal{H}_{\mathrm{SAG}}^{\mathrm{NN}}$. The prediction model is used to simulate n_e data instants by a disturbance estimator module to determine the value of ξ_k, described in Section 3.1.4. Then, the model is used with the calculated disturbance to simulate N data instants in the future by the optimization module described in Section 3.2.

3.1.1. Expert control model ($\mathcal{H}_{\mathcal{SAG}}^{\mathcal{EXP}}$)

The expert control system used in this study is based on plant operation knowledge through a fuzzy-logic framework combined with event-based rule sets. The fuzzy-logic framework maps each of the six CV-derived signals (values, rates, and accelerations) onto qualitative states such as *Very High*, *High*, *Ok* or *Low* using membership functions. Note that the state Very Low was not included, as low-pressure situations are not critical for mill operation and are already handled by the baseline control logic. The combination of these fuzzy values and their temporal patterns activate specific operational states, which are ordered hierarchically. The system distinguishes between stable operation (e.g., *Pressure Ok*) and riskier states (e.g., *High Pressure* or *Critical*) based on both the current values and recent trends of pressure and power. For each operational state, predefined rules define how each MV should be adjusted. These adjustments are implemented via the Sugeno method for fuzzy inference (Sugeno, 1985), where the strength of each MV move is determined by fuzzy rules and a set of configurable minimum and maximum step sizes.

Table 1 shows the critical states and the possible setpoint changes made by the expert control. Each row in the table is triggered by a specific condition:

- **Overload:** The pressure fuzzy state is *High* with its value increasing for 15 s and the power is decreasing for more than 5 s. This triggers in parallel a strong decrease in solids percentage, strong increase in speed, and a strong decrease in feed tonnage setpoints.
- **Power Critical:** The power exceeds its high limit and its rate of change fuzzy state is stable or increasing or when power fuzzy state is *Ok* or *High* and its value is increasing rapidly. The expert controller in these scenarios will increase strongly the solids percentage, softly decrease the speed, and strongly decrease the tonnage feed setpoints.
- **Pressure Critical:** Pressure alone exceeds the limit. The controller will in parallel reduce the solid percentage, increase the rotation speed, and also reduce the tonnage setpoint. The changes will be strong, regular or soft, depending whether the pressure is, respectively, *increasing fast*, *increasing not fast* or stable, according to the fuzzy states of its rate of change.
- **High Pressure states:** These states depend on pressure being at its *High* state, then will evaluate the condition of its rate of change as *increasing* or *not increasing* during 10 s, while also considering whether the power is at *High* state. For instance:
  - If *pressure is high and increasing, and power is high*, it suggests a regular decrease in solids percentage accompanied by, a strong or regular, decrease in tonnage feed, depending on the power values and rate of change.
  - If *pressure is high and increasing, and power is not high*, the controller suggests a decrease in the solid percentage, accompanied by a soft decrease in tonnage feed. The speed of

rotation may be increased or decreased, depending on the fuzzy states of the power and its rate of change.

– If *pressure is high and not increasing, and power is high*, the controller may suggest a soft decrease or soft increase in the solid percentage, depending if the power is increasing or stable, respectively. Independently from this, the controller will also suggest a soft increase in speed, and a strong or regular decrease in tonnage feed, depending on the pressure's rate of change.
– If *pressure is high and not increasing, and power is not high*, the controller suggests a soft decrease in the solid percentage, while also a soft increase in the speed of rotation. Lastly, it will suggest a strong or regular decrease in tonnage feed, depending on the pressure's rate of change.

The expert control system uses current and past values of the CVs and MVs to determine the next setpoint $\boldsymbol{u}^{SP}_{k+i+1}$ to be implemented. Specifically, for each sample k, the system constructs lag matrices of length p_y for CVs, p_{SP} for MV setpoints, and p_u for actual MVs. These matrices are used to compute the rate of change and acceleration of the CVs. The model in closed-loop is represented by Eq. (1), where counter i is used to represent the simulation's discrete time.

$$\hat{\boldsymbol{u}}^{SP}_{k+i+1} = \mathcal{H}^{\mathrm{EXP}}_{\mathrm{SAG}}\left(\boldsymbol{Y}^{\mathrm{MEAS}}_{k,p_y,i}, \boldsymbol{U}^{SP}_{k,p_{\mathrm{SP}},i}, \boldsymbol{U}_{k,p_u,i}, \boldsymbol{y}^{LIM}_{k+i}\right), \tag{1}$$

where:

- $\boldsymbol{Y}^{\mathrm{MEAS}}_{k,p_y,i} := \left[\hat{\boldsymbol{y}}^{\mathrm{MEAS}}_{k+i|k}, \dots, \hat{\boldsymbol{y}}^{\mathrm{MEAS}}_{k+1|k}, \boldsymbol{y}^{\mathrm{MEAS}}_{k}, \dots, \boldsymbol{y}^{\mathrm{MEAS}}_{k-(p_y-i-1)}\right] \in \mathbb{R}^{n_y \times p_y}$ is the matrix containing historical and predicted measurements of the CV over a window of size p_y,
- $\boldsymbol{U}^{SP}_{k,p_{\mathrm{SP}},i} := \left[\hat{\boldsymbol{u}}^{SP}_{k+i|k}, \dots, \hat{\boldsymbol{u}}^{SP}_{k+1|k}, \boldsymbol{u}^{SP}_{k}, \dots, \boldsymbol{u}^{SP}_{k-(p_{\mathrm{SP}}-i-1)}\right] \in \mathbb{R}^{n_u \times p_{\mathrm{SP}}}$ is the matrix of the predicted, current and historical MV setpoints over a window of size p_{SP},
- $\boldsymbol{U}_{k,p_u,i} := \left[\hat{\boldsymbol{u}}_{k+i|k}, \dots, \hat{\boldsymbol{u}}_{k+1|k}, \boldsymbol{u}_k, \dots, \boldsymbol{u}_{k-(p_u-i-1)}\right] \in \mathbb{R}^{n_u \times p_u}$ is the matrix of the actual MVs, comprising predicted, current, and past values over a window of size p_u.
- $\boldsymbol{y}^{LIM}_{k-t,n_e}$ represents the vector of high limits of the CVs.

In Eq. (1), for each prediction of the setpoints $\hat{\boldsymbol{u}}^{SP}$, the previous values of $\boldsymbol{u}$ and $\boldsymbol{y}^{\mathrm{MEAS}}$ are needed. These values are provided either as real historical data (when $i = 0$) or as predictions (when $i > 0$). Additionally, the limits $\boldsymbol{y}^{LIM}_k$ and the bounds $\boldsymbol{u}^{SP}_k|^{min}_{max}$ are considered constant throughout the prediction horizon. As a consequence of these bounds, it is important to distinguish between saturation and optimality in the manipulated variables. When the expert controller stops adjusting a specific MV, such as speed or density, this often indicates that the variable has reached a predefined bound rather than an optimal operating point. These bounds, typically imposed by plant operators for safety or mechanical constraints, act as hard limits on MV behavior. Once an MV is saturated, the controller focuses on other available variables to maintain control of the CVs. Currently, operators and supervisors rely on manual adjustments of $\boldsymbol{y}^{LIM}$ to define the desired operational area. Poorly defined $\boldsymbol{y}^{LIM}$ can hinder the controller's response to disturbances. One of the motivations for introducing the D-RTO layer in this work is precisely to enable automatic and adaptive tuning of these limits to improve response to critical states.

3.1.2. Regulatory control model ($\mathcal{H}^{PID}_{SAG}$)

The performance of the PID loops affects the predicted operation of the SAG mill model, as the system's dynamics rely on the regulatory control layer. Specifically, the setpoint values $\hat{\boldsymbol{u}}^{SP}_{k+i|k+1}$ determined by the expert control system may differ from the process values $\boldsymbol{u}_{k+i|k}$ due to the response time of the loops. To address this, a state-space representation for each PID loop was developed and adjusted using historical data. The state-space models accurately simulate the dynamics of the PID loops for tonnage feed, mill rotation speed, and solids percentage,

Table 2
Results of accuracy and error description for the state-space models of tonnage, speed, and density.

Model	Average error	RMSE	R Value
Tonnage	$8.4 \times 10^{-3} \pm 0.7\%$	18.3 tph	0.97466
Speed	$-1.6 \times 10^{-5} \pm 0.08\%$	0.1%	0.99955
Density	$2.0 \times 10^{-4} \pm 0.2\%$	0.1%	0.99512

with initial values for the state variables estimated for each discrete simulation instant. Using the *System Identification Tool* in MATLAB, PID models were fit to data, and validated for each control loop. The sampling time was resampled to 30 s for consistency with the prediction horizon.

At each discrete instant k, the expert control system determines the setpoints $\hat{u}^{SP}_{k+1}$ for the next instant $k+1$ using Eq. (1). Then, Eq. (2) is used as a representation of the PID loops that will estimate the value of $\hat{\boldsymbol{u}}_{k+i+1|k}$ as follows:

$$\hat{\boldsymbol{u}}_{k+i+1|k} = \mathcal{H}^{\mathrm{PID}}_{\mathrm{SAG}}\left(\boldsymbol{U}_{k,\ell,i}, \boldsymbol{U}^{SP}_{k,\ell,i}, \varepsilon, x_{k=-\ell}\right), \quad i = 0, \dots, N-1. \tag{2}$$

where ε is a constant additive error, and ℓ is the order of the model. Note that:

- For $i = 0$, real historical values of $\boldsymbol{u}_{k,\ell}$ and $\boldsymbol{u}^{SP}_{k,\ell}$ are used.
- As i increases, predicted values $\hat{\boldsymbol{u}}_{k+i}$ are used until the end of the prediction horizon N.

The behavior of the regulatory PID loops was identified from plant data in order to capture anti-windup actions, actuator limits, and nonlinearities that are not always reflected in the nominal tuning. This approach ensures that the closed-loop dynamics represented in the digital twin are consistent with the actual regulatory performance. The error analysis results for the PID models are summarized in Table 2, with detailed predictions and error profiles provided in Figures S1 to S3 of the Supplementary Material.

3.1.3. SAG model ($\mathcal{H}^{\mathcal{NN}}_{SAG}$)

The complex dynamics of the SAG mill are difficult to model as several factors affect the movement of material inside the mill, including size distribution, rheology, and ore composition. To address this, a recursive neural network (RNN) model is employed to simulate the process as a black-box system using the available information. The RNN can be trained to capture the complex dynamics of the process, and the best fit of the model is obtained through optimization. The RNN was trained and validated with the *Neural Net Time Series* toolkit in MATLAB®, providing an accurate description of the effects of the inputs $\boldsymbol{u}$ on the outputs $\boldsymbol{y}$. The dataset was acquired over 5-s intervals and subsequently filtered to a 30-s interval using the following operational conditions: SAG mill run signal ON; Tonnage feed above 1800 tph; Solids percentage above 72%; and Expert control ON.

These conditions represent normal operation, excluding startup or shutdown data. A total of 68 h of continuous operation, corresponding to 8156 data points, were collected. The dataset was split into training, validation, and testing subsets. Various configurations of the RNN were tested to determine the best performance. The final model, optimized using the *Levenberg–Marquardt* algorithm in MATLAB, consisted of one hidden layer, two neurons per neural net, and a 12-point data window. The optimization process terminated when the Jacobian of the performance value no longer improved as the adaptive value increased. The performance of the best-performing RNN models for mill pressure and power draw is summarized in Table 3, with detailed predictions and error profiles provided in Figures S4 and S5 of the Supplementary Material.

The resulting RNN model for the SAG mill is represented as:

$$\hat{y}_{k+i+1|k} = \mathcal{H}^{\mathrm{NN}}_{\mathrm{SAG}}\left(\boldsymbol{Y}_{k,m,i}, \boldsymbol{U}_{k,m,i}\right), \quad i = 1, \dots, N \tag{3}$$

Table 3
Results of accuracy and error description of the trained RNN for pressure and power draw of the SAG mill.

Model	Avg error	RMSE	R Value
Pressure	$-1.3 \times 10^{-5} \pm 0.1\%$	0.4 psi	0.99988
Power	$5.5 \times 10^{-4} \pm 0.2\%$	15.5 kW	0.99987

where:

- $\boldsymbol{Y}_{k,m,i} := \left[\hat{\boldsymbol{y}}_{k+i|k}, \dots, \hat{\boldsymbol{y}}_{k+1|k}, \boldsymbol{y}_k, \dots, \boldsymbol{y}_{k-(m-i-1)}\right] \in \mathbb{R}^{n_y \times (m+1)}$ is the matrix of the predicted, current, and past measurements CVs over a window size of $m+1$,
- $\boldsymbol{U}_{k,m,i} := \left[\hat{\boldsymbol{u}}_{k+i|k}, \dots, \hat{\boldsymbol{u}}_{k+1|k}, \boldsymbol{u}_k, \dots, \boldsymbol{u}_{k-(m-i-1)}\right] \in \mathbb{R}^{n_u \times (m+1)}$ is the matrix of the predicted, current and past MVs over a window size of $m+1$.

Here, $\boldsymbol{Y}_{k,m,i}$ and $\boldsymbol{U}_{k,m,i}$ represent the past $m = 12$ values of the controlled variables (CVs) and manipulated variables (MVs), respectively, required to predict $\hat{\boldsymbol{y}}_{k+i+1|k}$. At the first step ($i = 1$), only historical data is used, while subsequent steps use feedback predictions of $\boldsymbol{y}$ and $\boldsymbol{u}$.

3.1.4. Disturbance model ($\mathcal{H}_{dist}$)

To account for the plant-model mismatch between the predictions and actual process measurements, a disturbance estimation module is employed. This disturbance estimator uses known past values of the measured controlled variables $\boldsymbol{Y}^{\text{MEAS}}_{k-t}$, the setpoints $\boldsymbol{U}^{\text{SP}}_{k-t}$, and the manipulated variables $\boldsymbol{U}_{k-t}$ to estimate the disturbance vector $\boldsymbol{\xi}_{k,n_e}$. This vector minimizes the prediction error between the estimated and measured values.

The disturbance vector is defined as:

$$\xi_{k,n_e} = \left[\xi_{k-n_e+1|k}, \dots, \xi_{k-1|k}, \xi_{k|k}\right]. \tag{4}$$

Here, n_e is the number of discrete time instants prior to k used for the estimation.

The disturbance is modeled as additive to the predicted process outputs:

$$\hat{\boldsymbol{Y}}^{\text{MEAS}}_{k-t|k} = \hat{\boldsymbol{Y}}_{k-t|k} + \boldsymbol{\xi}_{k-t}, \tag{5}$$

where $\hat{\boldsymbol{Y}}_{k-t|k}$ represents the model predictions at time $k-t$.

The disturbance ξ_k is assumed constant throughout the estimation horizon n_e, such that:

$$\forall t = 0, \dots, n_e - 1 \;:\; \xi_{k-t} = \xi_k. \tag{6}$$

The constant additive disturbance across the prediction horizon which is based on the offset-free model predictive control framework. This framework is a standard approach to compensate for plant–model mismatch without steady-state offset (Muske & Badgwell, 2002; Pannocchia & Rawlings, 2003).

The disturbance estimation problem is formulated as a minimization of the squared prediction error:

$$\min_{\xi_k} \sum_{t=0}^{n_e-1} \left(\hat{\boldsymbol{Y}}^{\text{MEAS}}_{k-t|k} - \boldsymbol{Y}^{\text{MEAS}}_{k-t}\right)^2 \tag{7}$$

s.t.:

$$\mathcal{H}\left(\boldsymbol{Y}^{\text{MEAS}}_{k-t,n_e+m}, \boldsymbol{U}^{\text{SP}}_{k,n_e+m}, \boldsymbol{U}_{k,n_e+m}, \boldsymbol{y}^{LIM}_{k-t,n_e}, \boldsymbol{\xi}_k\right) = 0,$$

where:

- $\boldsymbol{Y}^{\text{MEAS}}_{k-t,n_e+m} \in \mathbb{R}^{n_y \times (n_e+m)}$ is the matrix of historical and predicted measurements of the CVs. It includes the last n_e historical measurements and m predicted future values, forming a total window of $n_e + m$ data points.
- $\boldsymbol{U}^{\text{SP}}_{k,n_e+m} \in \mathbb{R}^{n_u \times (n_e+m)}$ is the matrix of setpoints for the MVs. It includes the last n_e historical setpoints and m predicted future setpoints, forming a total window of $n_e + m$ data points.
- $\boldsymbol{U}_{k,n_e+m} \in \mathbb{R}^{n_u \times (n_e+m)}$ is the matrix of historical and predicted values of the MVs. It includes the last n_e historical values and m predicted future values, forming a total window of $n_e + m$ data points.
- $\boldsymbol{y}^{LIM}_{k-t,n_e}$ represents the vector of high limits of the CVs over the last n_e time instants. The window includes only historical data of size n_e.
- $\boldsymbol{\xi}_k$ is the constant disturbance to be estimated.

The disturbance estimator provides a constant disturbance term $\boldsymbol{\xi}_k$. The resulting prediction model disturbance can be expressed as:

$$\hat{\boldsymbol{y}}^{\text{MEAS}}_{k+i+1|k} = \mathcal{H}_{\text{dist}}\left(\hat{\boldsymbol{y}}_{k+i+1|k}, \boldsymbol{\xi}_k\right) = \hat{\boldsymbol{y}}_{k+i+1|k} + \boldsymbol{\xi}_k, \quad i = 1 \dots N \tag{8}$$

where $\hat{\boldsymbol{y}}_{k+i+1|k}$ is the disturbance-free prediction given by Eq. (3).

Eq. (8) uses the additive disturbance $\boldsymbol{\xi}_k$, which is obtained from the disturbance estimation module. As noted in Section 3.1.2, the expert control system uses the disturbed prediction $\hat{\boldsymbol{y}}^{\text{MEAS}}_{k+i+1|k}$ to define setpoints. This adjustment ensures that plant-model mismatches do not compromise the expert control's ability to stabilize the process.

3.2. Optimization module

The optimization searches the value of $\boldsymbol{y}^{LIM}$ to minimize an adequate objective function $\mathcal{F}$ as follows:

$$\begin{aligned}&\min_{\boldsymbol{y}^{LIM} \in \boldsymbol{y}^{LIM}} \mathcal{F}\left(\hat{\boldsymbol{Y}}^{MEAS}_{k,N}, \hat{\boldsymbol{U}}^{SP}_{k,N}, \hat{\boldsymbol{U}}_{k,N}\right)\\ &\text{s.t:}\\ &\mathcal{H}\left(\boldsymbol{Y}^{MEAS}_{k,N+m}, \boldsymbol{U}^{SP}_{k,N+m}, \boldsymbol{U}_{k,N+m}, \boldsymbol{y}^{LIM}_{k,N}, \boldsymbol{\xi}_{k,N}\right) = 0,\end{aligned} \tag{9}$$

where:

$$\begin{aligned}
\hat{\boldsymbol{Y}}^{MEAS}_{k,N} &:= \left[\hat{\boldsymbol{y}}^{MEAS}_{k+N|k}, \dots, \hat{\boldsymbol{y}}^{MEAS}_{k+1|k}\right] \in \mathbb{R}^{n_y \times N},\\
\boldsymbol{Y}^{MEAS}_{k,N+m} &:= \left[\hat{\boldsymbol{y}}^{MEAS}_{k+N|k}, \dots, \hat{\boldsymbol{y}}^{MEAS}_{k+1|k}, \boldsymbol{y}^{MEAS}_{k}, \dots, \boldsymbol{y}^{MEAS}_{k-(m-1)}\right]\\
&\in \mathbb{R}^{n_y \times (N+m)},\\
\hat{\boldsymbol{U}}^{SP}_{k,N} &:= \left[\hat{\boldsymbol{u}}^{SP}_{k+N|k}, \dots, \hat{\boldsymbol{u}}^{SP}_{k+1|k}\right] \in \mathbb{R}^{n_u \times N},\\
\boldsymbol{U}^{SP}_{k,N+m} &:= \left[\hat{\boldsymbol{u}}^{SP}_{k+N|k}, \dots, \hat{\boldsymbol{u}}^{SP}_{k+1|k}, \boldsymbol{u}^{SP}_{k}, \dots, \boldsymbol{u}^{SP}_{k-(m-1)}\right]\\
&\in \mathbb{R}^{n_u \times (N+m)},\\
\boldsymbol{U}_{k,N} &:= \left[\boldsymbol{u}_{k+N|k}, \dots, \boldsymbol{u}_{k+1|k}\right] \in \mathbb{R}^{n_u \times N},\\
\boldsymbol{U}_{k,N+m} &:= \left[\hat{\boldsymbol{u}}_{k+N|k}, \dots, \hat{\boldsymbol{u}}_{k+1|k}, \boldsymbol{u}_{k}, \dots, \boldsymbol{u}_{k-(m-1)}\right]\\
&\in \mathbb{R}^{n_u \times (N+m)},\\
\boldsymbol{\xi}_{k,N} &:= \left[\xi_{k+1|k}, \dots, \xi_{k+N|k}\right] \in \mathbb{R}^{n_y \times N},\\
\boldsymbol{Y}^{LIM}_{k,N} &:= \left[y^{LIM}_{k+N|k}, \dots, y^{LIM}_{k|k}\right] \in \mathbb{R}^{n_y \times N}.
\end{aligned}$$

The symbol $\hat{\cdot}$ indicates that the variable is estimated using the available prediction model $\mathcal{H}$, N is the prediction horizon, and m is the number of previous data points needed for the model. Note that $y^{LIM}_{k+i+1|k}$ is considered constant in the prediction horizon, such that $\boldsymbol{y}^{LIM}_{k+i+1|k} = \boldsymbol{y}^{LIM}_{k+1|k} = \boldsymbol{y}^{LIM} \quad \forall i = 0, \dots, (N-1)$. The idea behind this is to find the maximum allowed pressure (high limit) that provides maximum tonnage throughput and avoids critical conditions.

The formulation of the objective function $\mathcal{F}$ is grounded in operational expertise, incorporating insights from experienced operators about the trade-offs between pressure limits and tonnage throughput, and is defined as:

$$\mathcal{F} = -\alpha_1 \cdot \frac{\boldsymbol{y}^{LIM}_1}{\boldsymbol{y}^{LIM}_{1\ \text{max-allowed}}} - \alpha_2 \cdot \sum_{t=0}^{5} \left(\frac{\hat{\boldsymbol{u}}^{\text{SP}}_{1,k+t|k}}{6 \cdot \boldsymbol{u}^{\text{SP,max}}_1}\right), \tag{10}$$

where α_1 is the weight of the pressure variable, α_2 is the weight of the tonnage throughput variable, $\boldsymbol{y}^{LIM}_1$ is the current high limit of pressure, $u^{\text{SP,max}}_1$ is the maximum high limit allowed for the setpoint, and $\boldsymbol{y}^{LIM}_{1\ \text{max-allowed}}$ is the boundary condition for optimization, defined

Algorithm 1: Search routine for the optimization module

Set : $num_{test} \leftarrow 20, \quad step_{min} \leftarrow 0.2$

Simulate the predictive model over the prediction horizon N using ξ_k from the disturbance estimator module, available data $\boldsymbol{Y}^{MEAS}_{k,m,i}$, $\boldsymbol{U}^{SP}_{k,m,i}$, $\boldsymbol{U}_{k,m,i}$, and the current y^{LIM}

if *at least one of the tonnage setpoints* $u^{SP}_1|_{k+i}, \forall i = 1, \ldots, N$ *is less than* u^{SPmax}_1 **then**

  Calculate $\mathcal{F}_{\text{value}}$

  Set : $\mathcal{F}_{\text{best}} \leftarrow \mathcal{F}_{\text{value}}, \quad y^{LIM}_{1,\text{best}} \leftarrow y^{LIM}_1, \quad y^{LIM}_{max-allowed}, l_b, l_{b_{int}}, \quad step_{\text{prev}} \leftarrow 0$

  $step_{\text{initial}} \leftarrow \frac{(y^{LIM}_{max-allowed} - l_b)}{num_{\text{test}}}$

  `// Outer-loop`

  for *the first loop run:* ***Set*** *Step* $\leftarrow step_{initial}$ **do**

    Then: Update step $\leftarrow step \times 0.6$

    if $step \leq step_{min}$ **then**

      $step \leftarrow step_{\text{min}}$

    if $step = step_{prev}$ **then**

      Stop outer loop

      Set $y^{LIM}_{1|k+1} \leftarrow y^{LIM}_{1,\text{best}}$

      Exit and go to wait next instant k

    $step_{\text{prev}} \leftarrow step$

    `// Inner-loop`

    for *the first loop run:* ***Set*** $y^{LIM}_1 \leftarrow y^{LIM}_{max-allowed}$ **do**

      Update $y^{LIM}_1 \leftarrow y^{LIM}_1 - step$

      if $y^{LIM}_1 \leq l_{b_{int}}$ **then**

        Stop inner-loop and **Go** to start outer-loop

      else

        Simulate the predictive model over the prediction horizon N with y^{LIM}_1

        Calculate current-value of obj fun $\mathcal{F}_{\text{value}}$

        if *the overload state is active at* $i = [1, 2]$ **then**

          Update $l_{b_{int}} \leftarrow \max\left(l_b, y^{LIM}_1\right)$

          Stop inner-loop and **Go** to start outer-loop

        else

          if *current-value* $\leq$ *best-value* **then**

            $\mathcal{F}_{\text{best}} \leftarrow \mathcal{F}_{\text{value}}$

            $y^{LIM}_{1,\text{best}} \leftarrow y^{LIM}_1$

            Go to start inner-loop

else

  Exit and go to wait for next instant k

as the maximum allowed high limit of pressure. The values for the weights α_1 and α_2 were found by trial and error, with values of 0.5 and 1, respectively.

A tailored algorithm was developed specifically for this problem, considering that the digital twin contains both continuous and discrete variables. To address this complexity, a customized optimization approach was designed, as detailed in Algorithm 1. This ad-hoc search routine leverages operational knowledge of the process and its control systems to navigate the optimization space effectively. The algorithm comprises two main loops: an outer loop and an inner loop. Together, these loops iteratively search for the optimal high pressure limit, y^{LIM}_1, that maximizes throughput while avoiding critical conditions such as overloading.

Outer loop: The outer loop is responsible for iteratively reducing the step size used in the search, starting with an initial value and progressively decreasing it by a factor of 0.6. This reduction balances the resolution of changes in the objective function $\mathcal{F}$ while preventing excessive iterations that would increase computational effort. The outer loop is triggered only when the simulation results with a prediction horizon $N = 6$ predict that the setpoint for tonnage, $u^{\text{SP}}_{1|k+i}$, $i = 1, \ldots, N$, is below its maximum allowed value for at least one step in the horizon. If this condition is met, the algorithm stores the initial objective function value ($\mathcal{F}_{\text{value}}$) and the corresponding high pressure limit (y^{LIM}_1). It then initializes the search parameters, including the initial step size ($step_{\text{initial}}$), the minimum allowable step size ($step_{\text{min}} = 0.2\,\text{psi}$), and the boundary conditions. The outer loop also ensures that if the step size falls below the minimum $step_{min}$, or if no further improvements are found, the search is terminated, and the best high-pressure limit, $y^{LIM}_{1,best}$ is applied.

Inner loop: The inner loop performs the search for the optimal value y^{LIM}_1 by iteratively evaluating the objective function $\mathcal{F}$ over the search space defined by the current bounds. Starting from the upper bound $y^{LIM}_{\text{max-allowed}}$, the algorithm reduces the high pressure limit (y^{LIM}_1) by the current step size in each iteration. The predictive model, with a horizon of $N = 6$, is simulated using the current y^{LIM}_1 and the disturbance estimate ξ_k. At each iteration, the algorithm evaluates the objective function value ($\mathcal{F}_{\text{value}}$) and compares it with the best value found so far ($\mathcal{F}_{\text{best}}$). If the current value improves upon the best value, it updates $\mathcal{F}_{\text{best}}$ and stores the corresponding y^{LIM}_1 as $y^{LIM}_{1,best}$.

If an overload condition occurs during the simulation, the inner loop is terminated, and the algorithm updates $l_{b_{int}}$ to ensure that subsequent iterations exclude the problematic region of the search space. The new $l_{b_{int}}$ is calculated as:

$$l_{b_{int}} = \max(l_b, y^{LIM}_1), \tag{11}$$

where l_b is the dynamic lower bound determined as the maximum of 95% of $y^{LIM}_{max-allowed}$ and 99% of the current pressure value, i.e.,:

$$l_b = \max\left(0.95 \cdot \mathbf{y}^{LIM}_{1max-allowed}, y^{LIM}_{1,k} \cdot 0.99\right) \tag{12}$$

If the current y^{LIM}_1 reaches $l_{b_{int}}$, the inner loop terminates, and the control returns to the outer loop to adjust the step size and restart the search from a higher solution.

The algorithm continues iterating through the outer and inner loops until either the step size reaches its minimum or no further improvements are found. If the search identifies a better high pressure limit (y^{LIM}_{1best}), it is applied to the process for the next discrete time interval. Conversely, if no improvement is detected, the D-RTO layer retains the current high limit without making any changes. This approach ensures that the D-RTO layer dynamically identifies the optimal high pressure limit, leveraging predictive capabilities to avoid critical operational states. This approach is non-intrusive for industrial implementation, preserves the established operator–expert interaction, and focuses the predictive layer only on the conditions where the expert is known to face limitations.

4. Results and discussions

The optimization strategy proposed in Section 3.2 was implemented in Python 3.7, with a sampling time of 30 s. The RNN used to model SAG mill dynamics is assumed to be the ground truth. Then, we add an external disturbance as expressed in Eq. (8). The disturbance model simulates the reaction to changes in operational conditions within the mill or variations in ore characteristics, leading to the expert system detecting *Critical* or *High Pressure* states when control actions are inadequate to avoid these states. An *Overloading* state can occur due to several factors, including a reduction in the milling capacity of the SAG mill, an increase in ore hardness, or issues related to the characteristics of the pulp and internal dynamics. These factors can negatively impact the ability to reduce ore size and the throughput of solids. These impacts can be seen as an increase in the pressure of the SAG mill hydraulics and, at a certain point, a decrease in the power draw of the mill. It is expected that before the *Critical* or *High Pressure* states are active, the D-RTO layer will perform the optimization routine described in Section 3.2. This will provide the expert control with a high limit pressure that changes and adapts to the operating conditions. Note that the disturbance affects the PID loops and the expert control layer (see Fig. 2). Consequently, the D-RTO layer uses measurements $\mathbf{y}^{MEAS}$, which are offset from the SAG model values by ξ_k, to determine actions and states. Modifying the pressure limit dynamically also seeks to avoid or quickly exit the *Critical* and *High Pressure* states whenever feasible. This indicates that the D-RTO layer must reduce the overall duration for which the expert control assesses these critical situations or, at least, decrease the frequency of their occurrence.

Three scenarios are simulated to assess the effectiveness of the proposed D-RTO strategy in reducing the effects of undesired states associated with tonnage loss. These scenarios consider two types of disturbances to emulate increased ore hardness at the inlet, and two configurations of $\mathbf{u}^{SP}|^{max}_{min}$ to evaluate how the availability of control resources influences system response. A summary of the conditions for each scenario is provided in Table 4. It should be noted that the bounds reported for mill speed (u_2) in Table 4 correspond to a pseudo-critical speed defined by the plant operations team. In the industrial plant considered, the SAG mill operates between 90% and 100% of the pseudo-critical speed. This pseudo-critical definition, adopted by the plant, accounts for liner wear and effective internal dimensions, and corresponds to approximately 90% of the theoretical critical speed. Therefore, the operating window of 90%–100% pseudo-critical speed is equivalent to 81%–90% of the theoretical critical speed. This range is consistent with reported values for industrial tumbling mills, where ball mills usually operate at 70%–80% of critical speed, and semi-autogenous mills may reach up to 90% under conditions where slippage between the load and the liners requires higher rotational speeds to achieve acceptable charge motion (Liddell & Moys, 1988; Powell, 1991).

In all cases, it is assumed that prior to the disturbance, the mill is operating under nominal conditions classified as '*OK*' by the expert control system, with maximum throughput and full availability of speed and density adjustments. Accordingly, the initial conditions are defined as $u_1 = u^{SP}_1|^{max}$, $u_2 = u^{SP}_2|_{min}$, and $u_3 = u^{SP}_3|^{max}$. The purpose of defining these initial conditions is twofold: first, to establish a reference point representing stable operation under expert control; and second, to enable the expert system to react appropriately to disturbances by leveraging the full control hierarchy described in Section 2, thereby avoiding overestimation of the benefits achieved through the proposed supervisory strategy.

Disturbances are defined to replicate the impact on pressure and power resulting from an increase in the ore hardness at the inlet. The profiles of these disturbances are described as follows:

- **Ramp disturbance:** Since pressure is proportional to the mill inventory, an additive ramp disturbance is used to mimic a reduction in the outflow caused by a sudden increase in ore hardness. Therefore, it is assumed that a step change in the ore hardness of the influent, leads to a steady increase in the pressure and a steady decrease in power draw.
- **Ramp + sinusoidal disturbance:** A sinusoidal component is added to the ramp pressure disturbance to simulate more complex scenarios, such as ore mixing or gradual changes in feed composition, which often occur due to the unpredictable behavior of the stockpile feeding the mill.

Fig. 3 illustrates the disturbances applied in each scenario. For pressure, the ramp disturbance is segmented into multiple periods with varying slopes, while the sinusoidal component has a defined amplitude of 4 and a period of 50. The RNN model for the SAG mill ($\mathcal{H}^{NN}_{SAG}$, see Section 3.1) requires 12 previous time steps to predict the next value. Therefore, the simulation is initialized using 12 historical data points to provide consistent initial conditions for all relevant variables that influence the prediction. After this initialization, the process model predicts the next discrete time instant, and the D-RTO layer forecasts $N = 6$ future steps at each iteration. Notice that the model of disturbance for $\mathcal{H}_{dist}$ assumes a constant value for ξ_k along the prediction horizon (see Section 3.1.4), which introduces a source of structural modeling-mismatch between the internal model of the supervisory layer and the process.

With the disturbances applied in each scenario, the simulated process is driven into a critical state as recognized by the expert control system, primarily triggered by the pressure increase. The expected reaction from the expert control is a series of aggressive setpoint adjustments, particularly a reduction in the tonnage feed rate. These actions typically follow other control adjustments that may prove insufficient to counteract the effect of the disturbances on the controlled variables.

The disturbances were deliberately restricted to scenarios that lead to overload conditions, since these are the operating regimes where the expert controller faces limitations and the D-RTO is expected to provide improvements. From an operational perspective, overload events are typically associated with ore hardness and inventory build-up, which manifest as an increase in bearing pressure followed by a reduction in power draw once slurry pooling develops (Golpayegani & Rezai, 2022). In contrast, soft ore conditions generally reduce the mill load and may result in lower pressure, but such situations are not critical, as they are routinely managed by the existing expert controller through its stabilization logic and feed rate adjustments. For this reason, soft ore scenarios were not included in the analysis. The disturbances considered were therefore specifically designed to evaluate the D-RTO's capability under adverse conditions, namely overload and critical inventory accumulation

Table 4
Conditions of the scenarios. Limits are constant at all times, except for the pressure high limit, which the D-RTO layer can change. Note: Mill speed (u_2) is normalized to a pseudo-critical speed, which is defined as 90% of the theoretical critical speed. Thus, a value of 100% in this context corresponds to 90% of the actual physical critical speed.

	y_1^{LIM} [psi]	y_2^{LIM} [kW]	$u_1^{SP}\|_{min}^{max}$ [tph]	$u_2^{SP}\|_{min}^{max}$ [%]	$u_3^{SP}\|_{min}^{max}$ [% Solids]	Disturbance
Scenario 1	980.0	14,500	1,500–3,000	90.0–100.0	73.0–77.0	Ramp
Scenario 2	980.0	14,500	1,500–3,000	90.0–100.0	73.0–77.0	Ramp + Sinusoidal
Scenario 3	980.0	14,500	1,500–2,500	86.0–100.0	73.0–76.0	Ramp

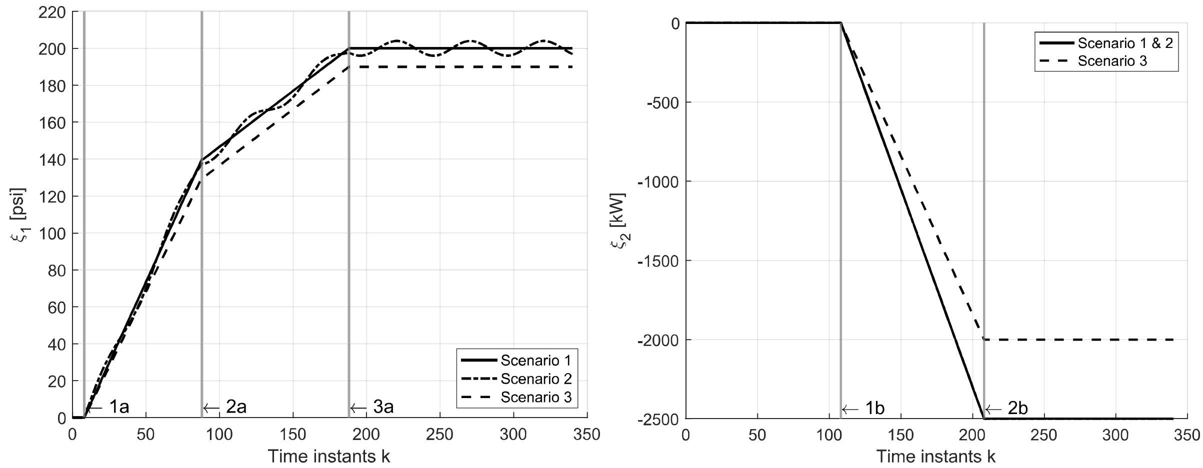

Fig. 3. Disturbance values for each scenario. (a) Pressure disturbances (ξ_1). Continuous line for scenario 1, dot dashed line for scenario 2, and dashed line for scenario 3. (b) Power disturbances (ξ_2). Continuous line for scenario 1 and 2 and dashed line for scenario 3. Gray vertical lines define limits for periods with different slopes of the disturbances, which are identified as 1a, 2a, and 3a for pressure disturbances; and 1b and 2b for power disturbances.

For clarity and brevity, only scenario 1 is presented in full detail in this section as a representative case. The outcomes of scenarios 2 and 3 are summarized in Table 5, while the plots, results analyses, and discussions are provided in the Supplementary Material.

In scenario 1, the SAG mill was subjected to a ramp disturbance simulating a sudden increase in ore hardness at the inlet, causing changes in pressure and power draw within the mill. The pressure disturbance began from zero and increased to 140 psi over 80 discrete time intervals, rising to 200 psi over the next 100 intervals. A linear reduction in power draw occurred, reaching a decrease of 2500 kW at $k = 208$, after which the disturbance values were maintained constant. These disturbances are summarized as follows:

- Pressure increases from $\xi_1 = 0$ at $k = 8$ to $\xi_1 = 140$ at $k = 88$ (period between 1a to 2a in Fig. 4(a) and Fig. 4(b)).
- Pressure increases further from $\xi_1 = 140$ to $\xi_1 = 200$ at $k = 188$ (period between 2a to 3a inFig. 4(a) and Fig. 4(b))
- Power decreases linearly, reaching $\xi_2 = -2500$ at $k = 208$. (period between 1b to 2b in Fig. 4(c) to Fig. 4(d))

Fig. 4 illustrates the effect of disturbances on the pressure and power profiles for scenario 1, comparing the simulation with and without D-RTO. In Fig. 4(a) and Fig. 4(b), the measured pressure (solid black line) increases steadily due to the ramp disturbance, while the modeled value without disturbances (dashed gray line) remains constant. The divergence between these curves highlights the impact of the external disturbance on system behavior. Fig. 4(c) and Fig. 4(d) show the power profile under both control strategies. A noticeable difference is observed: while both cases experience a drop in power due to the applied disturbance, the case with D-RTO exhibits smoother responses. This is attributed to the optimization strategy, which adapts the pressure limits and mitigates abrupt variations in throughput and load.

Fig. 5 presents the variations in pressure, tonnage, speed, and density under both control conditions, highlighting critical operational states. The *Overload* state is represented by red shaded regions, and other *Critical* or *High Pressure* states are represented by yellow shaded regions (see Table 1). There are four distinct periods labeled A, B, C, and D, each with subscripts I (D-RTO inactive), and A (D-RTO active). The D-RTO seeks to dynamically modify the upper pressure limit so that the actions of the expert controller become more aggressive. This adjustment is made under the assumption that the maximum achievable pressure is lower than the static high limit defined by the operator. In periods where the other manipulated variables are saturated, maintaining the operator-defined limit unchanged may lead to overload, since the expert controller's actions would be too conservative to prevent entering a critical state. Each period of scenario 1 is analyzed below:

Periods A_I and A_A: The first *Overload* state is triggered in both A_I and A_A at $k = 150$, lasting until $k = 160$. As shown in Fig. 5(c) and Fig. 5(d), the expert control system reduces the tonnage feed setpoint to $u_1^{SP} = 2828$ tph in both cases. No differences are noted during the initial overload condition. Although optimization is continuously executed, the D-RTO layer does not provide the expert control with an improved y_1^{LIM} value to apply. The overload is likely caused more by the rate of change in pressure rather than the proximity of the actual value y_1 to its limit y_1^{LIM}. The rapid increase in pressure results from the disturbance, and while the expert control executes expected actions regarding the mill's speed and density, these measures are insufficient to mitigate the rate of pressure change. Eventually, once the overload is no longer active, the tonnage setpoint is increased in both cases to $u_1^{SP} = 2880$ tph at $k = 162$, and then to $u_1^{SP} = 2930$ tph at $k = 165$.

Periods B_I and B_A: At time step $k = 170$, the D-RTO layer adjusts the pressure high limit to $y_1^{LIM} = 970.1$ psi. This preventive modification is designed to reduce predicted losses in tonnage throughput and is applied to the expert control system. As a result, the *High Pressure* state is triggered at $k = 171$, prompting the expert system to reduce the feed setpoint to $u_1^{SP} = 2871$ tph. By $k = 172$, an *Overload* state is detected in both B_A and inactive B_I cases. However, due to internal rules of the expert controller, including a minimum waiting time between successive MV changes, no further adjustment is made

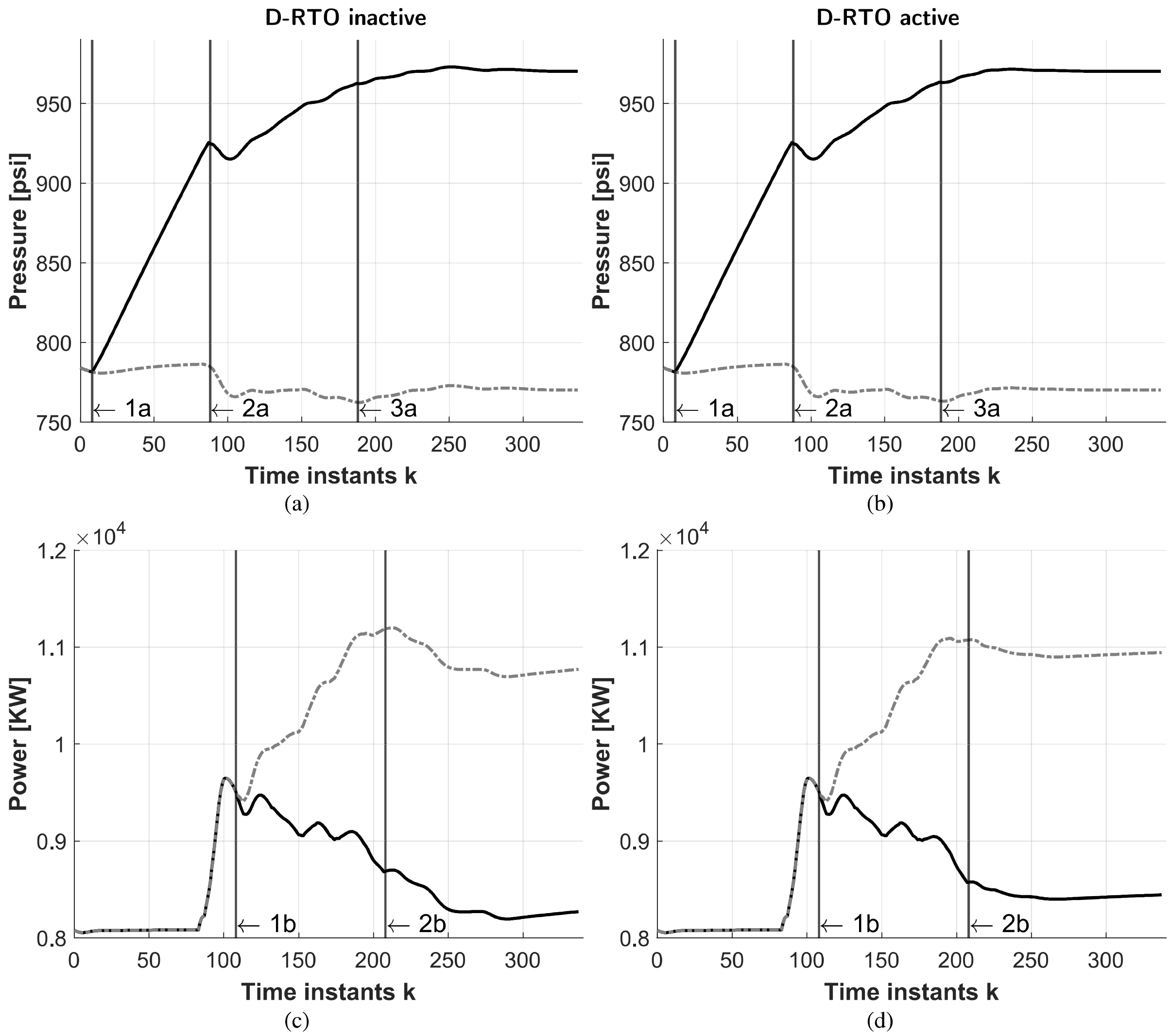

Fig. 4. Comparison of pressure (top) and power (bottom) for scenario 1 without (left) and with (right) D-RTO. Solid black lines show measured variables with disturbances ($\boldsymbol{y^{MEAS}}$), and dashed gray lines represent the modeled variables without disturbances ($\boldsymbol{y}$). Vertical lines (1a, 2a, 3a for pressure; 1b,2b for power) indicate disturbance applications points (see Fig. 3).

to the feed setpoint in $\mathrm{B_A}$. Additionally, the slowing rate of pressure increase at this point further delays any subsequent action.

In contrast, the scenario $\mathrm{B_I}$ maintains a higher pressure limit. This delays the activation of critical pressure-related states until the process is deeper into a critical region. Consequently, when the expert system finally detects an *Overload* state, it responds with a stronger reduction in feed, dropping the setpoint to $u_1^{SP} = 2805$ tph. This more aggressive action occurs both because the deviation from normal operation is more severe and because the internal timing constraint has elapsed. Thus, the setpoint trajectory differs across scenarios due to the indirect modulation of when and how the expert controller intervenes.

It should be noted that, although the trajectories of pressure and power appear visually similar between scenarios $\mathrm{B_I}$ and $\mathrm{B_A}$ during the first 220 samples, this is consistent with the relatively small difference in feed tonnage achieved by the D-RTO, which improves throughput by approximately 0.7% in this case. These differences are numerically significant and influence the timing and magnitude of actions taken by the expert controller.

Between periods $\mathbf{B_A}$ and $\mathbf{C_A}$: At $k = 191$, the D-RTO layer applies a new limit of pressure $y_1^{LIM} = 980.0$ psi. During the next 5 time instants, consecutive changes are applied. These are small changes that cause no significant effect in the setpoints and set the limit between $y_1^{LIM} = 978.9$ psi and 979.1 psi. This is the case until $k = 195$, when the change in y_1^{LIM} is accompanied by a setpoint change of $u_1^{SP} = 2915$ tph. Meanwhile, at the same discrete time k in the scenario with inactive D-RTO, the changes in the MVs are $u_1^{SP} = 2850$ tph.

Periods $\mathbf{C_I}$ and $\mathbf{C_A}$: The next *Overload* state is evaluated at $k = 198$, when the D-RTO layer has lowered the pressure limit to $y_1^{LIM} = 965.0$ psi and tonnage has been raised to $u_1^{sp} = 2976$ tph. It can be seen in Fig. 5(a) that the tonnage feed setpoint decreases to $u_1^{SP} = 2701$ tph, and the *Overload* state lasts for the equivalent of 8 min of operation. However, when the D-RTO layer is active (Fig. 5(b)), and *Overload* state occurs for the equivalent of only 1 min of operation, and then a *Critical/High pressure* state is active. At discrete time $k = 198$, the expert control changes the setpoint of the speed of rotation of the SAG mill to $u_2^{SP} = 99.73\%$ when the D-RTO layer is active. Without D-RTO layer, the setpoint is set at $u_2^{SP} = 99.66\%$. This is the first time that the speed setpoint is different for each case. The difference in speed setpoint change is likely to be caused by a difference in the rate of change of the pressure, as it is affected by the higher setpoint on feed tonnage, along with the difference in y_1^{LIM}. Note that the expert control sets a higher criticality of the state; therefore, changes are more aggressive when y_1 is closer to its limit.

At $k = 200$, the speed and solids percentage setpoints are already at their maximum and minimum, respectively, so no further action on these MVs is possible. The preventive lower tonnage action performed

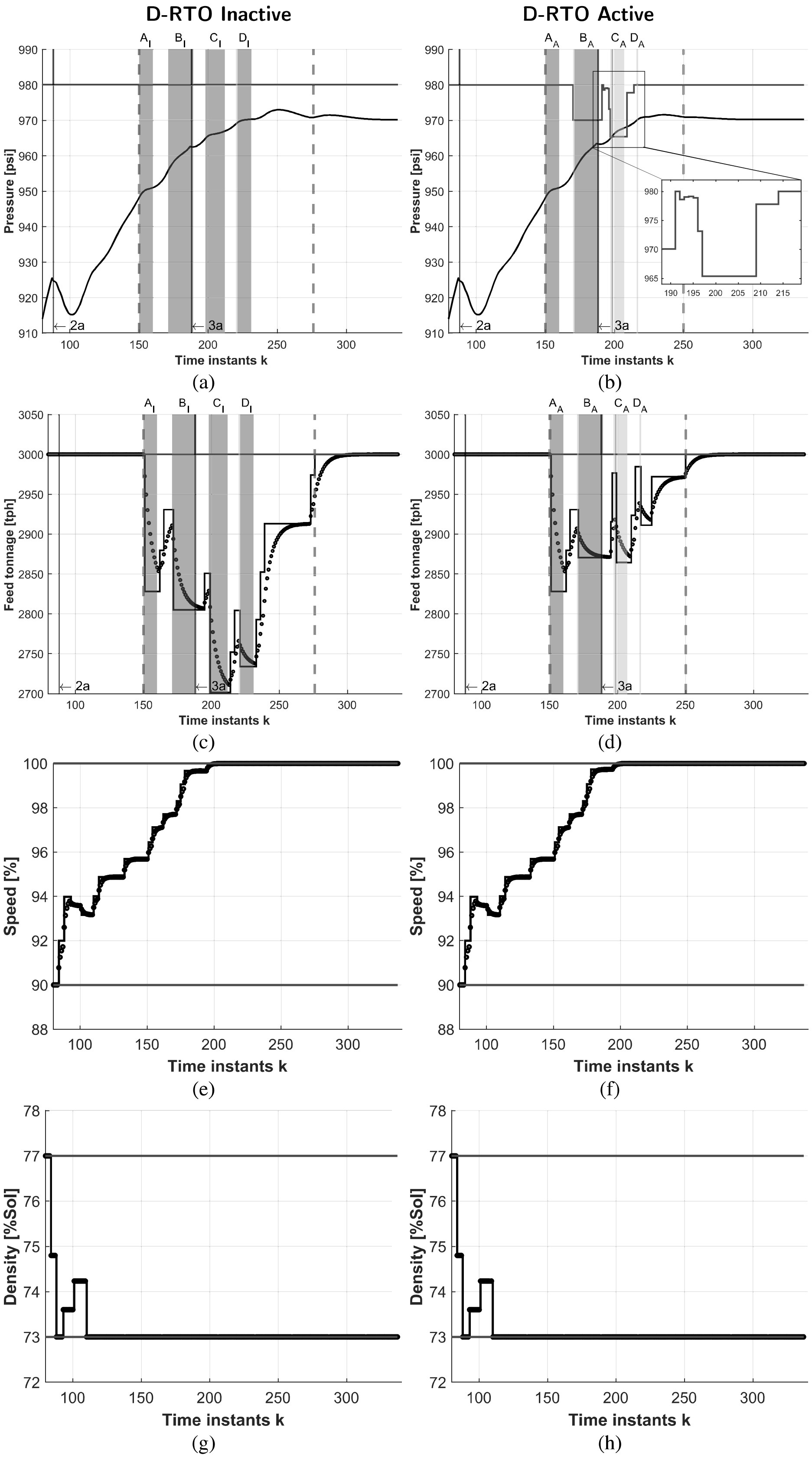

Fig. 5. Pressure and manipulated variables for scenario 1. Figures (a) and (b): black lines are the measured pressure y_1^{MEAS}, and green lines are the high limits y_1^{LIM}. Figures (c) to (h): black lines are the setpoints $\boldsymbol{u}^{SP}$, black dots are the value of the process variables $\boldsymbol{u}$, green lines are the limits $\boldsymbol{u}^{SP}|_{min}^{max}$. Red dashed lines indicate the time when tonnage is not at its maximum, and gray vertical lines indicate the disturbance applied for pressure (2a, 3a). Shades: red is for *Overload* and yellow for other *Critical* or *High Pressure* states (see Table 1).

at time $k = 171$ and other adjustments to $\boldsymbol{y}_1^{LIM}$ have the desired effect of causing shorter *Critical* states and reducing the loss of tonnage throughput.

Periods $\mathbf{D_I}$ and $\mathbf{D_A}$: Without the D-RTO layer, another *Overload* state occurs again at $k = 221$ that remains active for 10 time instants, equivalent to 5 min, with a tonnage setpoint of $u_1^{SP} = 2733$ tph. The D-RTO layer, however, finds a better pressure limit of $y_1^{LIM} = 977.0$ psi at $k = 209$, and then $y_1^{LIM} = 980.0$ psi at $k = 214$. This last change causes the tonnage setpoint to go from $u_1^{SP} = 2984$ tph at $k = 214$ to $u_1^{SP} = 2911$ tph at $k = 217$ when a *High Pressure* state is active. No overload occurs with the D-RTO layer active, and the ore feed setpoint is back at its maximum by $k = 250$. In comparison, without the D-RTO layer, the tonnage feed setpoint is back at its maximum only by discrete time $k = 276$, as seen in Fig. 5(a).

Overall, the effectiveness of the D-RTO layer is evident in Fig. 5(a) and Fig. 5(b). From $k = 149$ to $k = 250$, a period covering 50.5 min of operation, 49.5 min involve a reduced tonnage setpoint. Compared to the 62.5 min of suboptimal operation without D-RTO layer, this represents a 21% reduction in throughput limitation. On average, supervisory adjustments result in a tonnage feed setpoint loss of 90 tph below the maximum, corresponding to an average setpoint of 2909 tph. In the same period, the average loss without a D-RTO layer is 188 tph, and the average feed is 2811 tph. These results represent a 52.1% reduction in the loss of feed, and a 3.5% increase in the average tonnage setpoint feed. The improvement presented changes slightly when taking into account the fact that the period of tonnage lower than maximum is longer when no D-RTO layer is present, as in this case, the tonnage is back at maximum at $k = 275$ instead of $k = 250$ when the limit has been optimized. Nevertheless, the total tonnage feed loss is 176 tons in the simulation, and this value is reduced by 57% to 76 tons when the D-RTO layer is active.

In total, the D-RTO layer performed 102 iterations and made 10 changes in the pressure high limit of y_1^{LIM} which reduced the activation of the critical states from 28.5 min without the D-RTO layer to 22 min when it is active, which corresponds to a 23% reduction in time. The importance of the D-RTO layer results is due to reducing the loss of tonnage feed setpoint by 45% from its value, and this reduction is present during the shorter time, further extending the importance of the adaptive pressure limit. This improvement may also be attributed to a stabilization in the tonnage feed setpoint, as smaller and less frequent setpoint changes are made, the dynamic of the pressure will be consequently stabilized as well. This finding is of significant importance for the control system proposed, as stabilization has been mentioned to be a common improvement of different control systems for milling processes (Bouffard, 2015; Zhou, Lu, Yuan, & Chai, 2016), proving that this system is in line with the expectations of similar control systems, and may further improve the results of expert systems implementation. At the same time, the overall improvement in average tonnage setpoint, while setpoint is lower than its maximum, is generally in line with the economic optimization of the control systems (Bouffard, 2015).

Although only scenario 1 is discussed in detail in this section, scenarios 2 and 3 provide additional evidence of the consistency of the proposed strategy. To facilitate the comparison across scenarios, we introduce the variable total tonnage throughput ($T_{total_{ton}}$), defined as the cumulative sum of the tonnage setpoint $\boldsymbol{u}_1^{SP}$ over a specified simulation period:

$$T_{total_{ton}} = \sum_{k=a}^{b} u_{1_k}^{SP} \Delta t \tag{13}$$

where Δt represents the time step, expressed in hours. The total tonnage throughput is expected to be higher when the pressure limits are adjusted with D-RTO compared to scenarios without such optimization layer.

Additionally, we define the tonnage throughput loss ($T_{total_{loss}}$) as the cumulative sum of the deviation between the setpoint $\boldsymbol{u}_1^{SP}$ and the maximum allowable setpoint $\boldsymbol{u}_1^{SP\max}$ at each time step. This loss is represented by the tonnage feed setpoint deficit, $\boldsymbol{u}_{1_{\text{loss}}}^{SP}$. The D-RTO is able to anticipate and reduce this loss. The relationship between the setpoint, its deficit, and the maximum allowable setpoint can be expressed as:

$$u_1^{SP\max} = u_{1_k}^{SP} + u_{1_k}^{SP_{\text{loss}}} \tag{14}$$

The total tonnage throughput loss is then calculated as:

$$T_{total_{loss}} = \sum_{k=a}^{b} u_{1_k}^{SP_{\text{loss}}} \Delta t \tag{15}$$

Given that each discrete time step in the simulation and optimization corresponds to 30 s, the calculation is straightforward with Δt appropriately converted to hours. This approach enables a clear comparison of the scenarios and quantifies the benefits of a D-RTO layer in minimizing throughput loss.

While scenario 1 shows the largest impact, with a 57% reduction in total tonnage loss, scenarios 2 and 3 also exhibit clear improvements in throughput losses (23% and 22%, respectively), accompanied by reductions in average setpoint loss and in the duration of critical states. These outcomes confirm that the D-RTO consistently enhances the performance of the expert controller, regardless of the nature of the disturbance, with the magnitude of improvement depending on the severity and dynamics of the perturbation.

Table 5
Differences obtained in the three scenarios for the main performance evaluations: total loss of tonnage fed, average setpoint (SP) of tonnage fed, average SP loss, total time with SP of tonnage below its maximum, period length of critical states, and total active time.

Comparison	Scenario 1	Scenario 2	Scenario 3
Total tonnage loss	−57%	−23%	−22%
Optimization period			
Avg SP tph	3.5%	0.7%	0.6%
Avg SP tph loss	−52.1%	−22.6%	−22.1%
Total time SP < max	−21%	15%	−18%
Critical states			
Period length	−16%	−1%	−51%
Total Active time	−23%	3%	−33%

Table 5 provides a comparative analysis of the three scenarios, highlighting the key performance metrics defined in Eqs. (13) and (15). Across all cases, the implementation of the D-RTO layer consistently results in a reduction of total tonnage loss caused by decreased feed. These reductions range from 22% in scenario 3 to a substantial 57% in scenario 1. The largest improvement in scenario 1 reflects the effectiveness of predictive pressure constraint management under structured disturbances, enabling optimal setpoint scheduling and minimizing unnecessary throughput reductions. Even in the presence of more challenging and variable disturbances in Scenarios 2 and 3, the D-RTO layer still proves beneficial by maintaining feed closer to its maximum permissible value and avoiding excessive conservatism in expert control responses.

The system also demonstrates the ability to maintain a higher average tonnage feed setpoint in all scenarios. Increases range from 0.6% in scenario 3 to 3.5% in scenario 1, with the highest gain again observed under the more predictable disturbance conditions. These results indicate that the D-RTO strategy can operate within safe limits while still pushing the process toward higher throughput targets, improving overall system productivity. The reductions in both setpoint loss and the total time operating below maximum throughput further validate the effectiveness of the control strategy. Scenario 1 again shows the best performance, with a 52.1% reduction in average setpoint loss and a 21% reduction in time below the maximum. Scenario 3 also performs well, showing a 22.1% and 20% reduction in these metrics, respectively. Although scenario 2 resulted in a smaller 9% decrease in time below maximum, the improvements in tonnage throughput and reduced total loss still demonstrate the system's value, even when perfect prediction of future disturbances is difficult.

Finally, regarding the control of critical operating conditions, the supervisory system effectively reduces both the number and duration of critical states in Scenarios 1 and 3. Scenario 3 sees the greatest benefit in this aspect, with a 51% reduction in critical state duration and a 33% reduction in total time spent in critical operation. In scenario 1, the time in critical states is also reduced by 23%. Although scenario 2 indicates a slight 3% increase in the duration of the critical state, this does not lead to a decrease in performance. In fact, the tonnage feed improves and the overall losses are reduced. This suggests that the system's control actions are more informed and that tolerances are better balanced, especially in response to the frequent oscillations caused by the sinusoidal disturbance.

In summary, while the magnitude of improvement varies with the nature of the disturbance, the D-RTO layer consistently enhances throughput stability, reduces feed losses, and improves system responsiveness during critical states. The largest benefits are observed in structured, ramp-type disturbances, while even in more dynamic or abrupt events, the control framework provides an effective layer of optimization.

5. Conclusions

This work presented a D-RTO strategy to improve the performance of the typical expert control systems found in SAG mill operations. The system integrates dynamic real-time optimization, enabling predictive adaptation of key operational variables—particularly the high pressure constraint (y_1^{LIM}) to manage and mitigate the effects of *Critical* and *High Pressure* states. The proposed method improves traditional control structures by creating a supervisory layer that adjusts to changes in the system.

Results under three disturbance scenarios demonstrate that the D-RTO strategy significantly reduces the occurrence, intensity, and duration of critical operating conditions. In all scenarios, the system achieved the dual objective of (1) increasing average tonnage throughput and (2) stabilizing process dynamics, leading to smoother transitions, reduced pressure oscillations, and fewer aggressive corrective actions from the expert control system. The proactive adjustment of the pressure limit proved particularly effective in preventing overloads and accelerating recovery to nominal operation, thereby contributing to more consistent plant performance and higher productivity.

Additionally, the D-RTO layer exhibited smoother responses and consistent improvements under highly dynamic conditions, such as those induced by sinusoidal disturbances. Even in scenarios where pressure trends are difficult to predict, the optimization routine managed to preserve high throughput while minimizing total feed loss. In cases of abrupt overload conditions, the system's ability to act on short prediction horizons helped mitigate their impact and improve recovery time.

It is worth noting that power consumption remains a critical aspect for further investigation. Although the supervisory strategy improves tonnage recovery, the increased use of mill rotational speed as a manipulated variable could raise overall energy demand. This trade-off suggests the need for a more holistic optimization framework that balances throughput maximization with energy efficiency. Future work will consider multi-objective formulations that incorporate energy costs directly into the objective function. Approaches such as hierarchical reinforcement learning or adaptive economic model predictive control could offer a viable path toward intelligent trade-off management in real time. Additionally, the incorporation of ore property estimation, such as grindability or breakage rates, may further enhance disturbance prediction and model accuracy. Overall, this work highlights the potential of integrating real-time optimization with expert control systems to achieve performance improvements that are both operationally relevant and scalable to industrial mineral processing applications.

CRediT authorship contribution statement

Cristobal Mancilla: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Rodrigo Bruna:** Supervision, Resources, Conceptualization. **Paulina Quintanilla:** Writing – review & editing, Visualization, Validation, Supervision, Formal analysis, Data curation, Conceptualization. **Daniel Navia:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.conengprac.2025.106589.

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